Towards Artificial Intelligence: A Neuro-Symbolic Cognitive Architecture

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Artificial General Intelligence (AGI)

A system capable of learning and reasoning across diverse cognitive domains at or above a human level (Goertzel, 2014)

Key Characteristics: -

- **Transfer Learning:** Applying knowledge from source domains to unrelated target domains
- **Cognitive Synergies:** The interaction of perception, memory, and reasoning
- **Agency:** Goal-directed behavior beyond immediate prompt-response loops

Comparative Analysis: Transformers vs. AGI Requirements

Feature	SOTA (Transformers/LLMs)	Target AGI Architecture
Reasoning Type	Probabilistic Token Prediction (system 1)	Causal & Logical Inference (System 2)
Memory	Finite Context Window (Transient)	Long-Term Episodic & Semantic Memory
Knowledge Update	Static Weights (Requires Retraining)	Online/Continuous Learning
Failure Mode	Hallucination / Confabulation	“I don’t know” / Query for info

Literature Review: Defining the AGI Horizon

Core Hypothesis:

AGI is not merely an extension of current technology but a qualitatively different pursuit: creating synthetic intelligence with human-level scope and generalization.

Distinction from Agentic AI

While Agentic AI focuses on autonomous, multi-step workflows to achieve specific goals, AGI aims for universal adaptability across unknown domains.

Operational Standards for AGI

The “Well-Educated Adult” Metric:

Defining AGI as a system capable of matching the cognitive proficiency and versatility of a competent human adult

The “Atomic” Standard:

A rigorous benchmark requiring systems to demonstrate "atomic" understanding, simulating and reasoning about phenomena without reliance on statistical correlations or human pre-training.

Architectural Approaches

- **Symbolic:** Logic-based, universal but brittle
- **Emergentist:** Neural networks, good at patterns but weak in logic
- **Hybrid (Cognitive Synergy):** Combining both for robustness [Goertzel, 2014]
- **Brain-Inspired:** Using Spiking Neural Networks (SNNs) for biological efficiency [Raman et al., 2025]

Current Status and Measurement

- **The "Jagged" Cognitive Profile:** Models like GPT-4 are "Savants" (High Knowledge) but "Amateurs" in Memory/Sensory tasks [Hendrycks et al., 2025]
- **Score:** GPT-4 achieves only 27% on the "Well-Educated Adult" AGI metric.
- **The Illusion of Generality:** Systems use workarounds (e.g., RAG, Context Windows) to mask deep cognitive deficits [Olsher, 2024]
- **Agentic AI:** A transition phase focusing on autonomous workflows, but still lacking true universal adaptability [Joshi, 2025]

Research Gaps: Cognitive Deficits

- **The Causality Gap:** Models predict correlations but cannot explain why, preventing true understanding [Olsher, 2024]
- **Memory & Continual Learning:** Lack of dynamic, experiential memory causes "Amnesia" (resetting after every session) [Raman et al., 2025]
- **The Logic-Learning Disconnect:** Neural models struggle to integrate explicit logical rules with probabilistic learning [Goertzel, 2014]

Research Gaps: Trust And Operations

- **The “Black Box” Problem:** Lack of real-time interpretability makes high-stakes autonomy unsafe [Joshi, 2025]
- **Future of Verification:** Current systems fail definitive tests (e.g., "Olsher Test") requiring provable safety and transparency [Olsher, 2024]
- **Conclusion:** A New Cognitive Architecture is required to bridge these gaps

Problem Statement

Review of the literature indicates that neither pure Symbolic AI (brittle) nor pure Connectionist AI (unreliable) is sufficient.

Proposed Research Problem:

"Current architectures lack a unified framework that integrates Explicit Logical Reasoning (for reliability) with Long-Term Vector Memory (for context) into Neural Models."

Addressing this is critical for reducing hallucination and enabling autonomous decision-making.

Proposed Solution: Neuro-Symbolic Cognitive Grid (NSCG)

Objective: A unified framework merging Neural flexibility, Symbolic reliability, and Infinite Context

- **Neural Perception (Subsymbolic):** Handles unstructured data (Vision/Language) using standard LLMs
- **Vector-Grounded Memory (Context):** A "Hippocampal" module for long-term episodic storage (Agentic RAG)
- **Explicit Reasoning (Symbolic):** A logic core using "Weight-Calculatism" to validate decisions

The Architecture

The Challenge: Connecting "fuzzy" neural embeddings with "crisp" logical rules

The Connector: Graph Neural Networks (GNN)

- Acts as the translator layer
- Maps neural features to specific nodes in a Knowledge Graph

The Storage: Logical Atoms in Vector DB

- Experiences stored as "Logical Atoms" (indivisible facts), not just tokens
- Allows the system to recall verified truths before generating answers

Operational Workflow

Perception & Retrieval:

- Input → Neural Encoder → Fetch Context from Vector Memory

Validation (The "Firewall"):

- Reasoning Engine checks the neural draft against logical rules
- Calculation: *Weight = Benefit * Probability*
- Action: If the draft violates safety rules, it is rejected

Execution:

- Approved logic is converted back to natural language
- Result: A fluent and logically sound response

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Thank You